

BEDHIAFI OKBA SOFTWARE ENGINEER

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🌐 [BEDHIAFI Okba](#)

📍 Tunisia, Ben Arous

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🔗 [Portfolio](#)



PROFESSIONAL SUMMARY

Full-stack software engineer graduating in Computer Engineering, experienced in building web and mobile applications with Next.js, NestJS, React Native, Angular, and PostgreSQL. Developed RAG-based AI assistants and cloud-connected access control systems integrated with physical hardware. Seeking a full-time software engineering role to build impactful products and contribute across the full development lifecycle.

EDUCATION

Engineering Degree in Computer Science, ESPRIT (Private School of Engineering) 2021 – 2026 | Tunisia

PROFESSIONAL EXPERIENCE

TRUSTGYM | Gym Management & Access Control Platform [🔗](#), (Software Engineer) 2026 | Full Time | Tunisia

- Built an end-to-end gym management ecosystem with separate interfaces for staff (admin/cashier) and members, plus access control and an AI assistant.
 - Delivered a web admin app for gyms, members, coaches, subscriptions, payments, doors/access, inventory, invoices, cash register, KPIs, and real-time dashboards; integrated an AI documentation assistant (RAG with OpenAI and pgvector).
 - Developed a member mobile app for authentication, session browsing, slot reservation, live occupancy, peak-hour analytics, and profile management.
 - Implemented access control via ZKTeco and Hikvision (Python APIs, NestJS bridge, PM2/ngrok deployment) door ENTRY/EXIT events feed live occupancy and capacity status for all gyms.
 - Designed a modular NestJS backend with role-based auth (admin/member JWT), WebSockets, and integrations with AWS, plus RAG-based AI assistant for staff.
- Tech Stack:** Next.js, React, React Native (Expo), NestJS, TypeORM, PostgreSQL, pgvector, OpenAI (RAG), TanStack Query, Socket.io, AWS S3/SES, Docker, Python (ZKTeco/Hikvision APIs), PM2

FTB | Tunisian Bridge Federation [🔗](#), (Full-stack Developer) 2024 | Freelance | Tunisia

- Developed and deployed a federation management platform handling users, clubs, licenses, and tournament operations.
 - Built secure authentication, role-based dashboards, and admin workflows, increasing digital engagement by 30%.
 - Delivered a responsive UI with integrated data visualization for federation insights.
- Tech Stack:** Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, TanStack Query, React Hook Form, Chart.js, AWS S3

KREATEK | Multi-Tenant E-Commerce Platform, (Full-stack Developer) 2023 | Freelance | Tunisia

- Engineered a modular e-commerce platform supporting multiple brands with a scalable multi-tenant architecture.
 - Delivered four dedicated interfaces for customers, administrators, vendors, and delivery agents.
 - Focused on maintainability, role separation, and extensible business workflows.
- Tech Stack:** React, shadcn/ui, PostgreSQL, Express.js, JWT.

TRUSTLINK | Access Control & HR Management Platform [🔗](#), (Engineering internship) 2023 (3 months) | Internship | Tunisia

- Designed and implemented a full-stack platform combining facility access control and HR operations management.
 - Developed modules for buildings/zones/doors management, organizational structure, employee scheduling, leave requests, and audit logging.
 - Deployed scalable backend services with modern DevOps practices.
- Tech Stack:** NestJS, TypeORM, PostgreSQL, Next.js, React, TanStack Query, Tailwind CSS, shadcn/ui, Docker, AWS S3

ACADEMIC PROJECTS

SKILLY - AI-Powered Skill Exchange Platform [🔗](#), (Web site) 2025

- Designed and built a skill marketplace for exchanging expertise through real-time collaboration and AI-assisted mentoring.
 - Implemented onboarding, skill listings, secure payments, notifications, feedback workflows, gamification, and progress analytics.
 - Integrated AI components for personalized learning guidance and automated recommendations.
- Tech Stack:** React, NestJS, Redis, Docker, MongoDB, shadcn/ui, TensorFlow

E.V.H - EV Charging Station Maintenance Assistant, (Multi-Platform) 2023

- Built a multi-platform solution supporting technicians in the installation, repair, and maintenance of electric vehicle charging stations.
 - Delivered guided step-by-step workflows, real-time diagnostic reporting, and troubleshooting tools to improve intervention efficiency.
 - Connected service management features with payment processing for operational support.
- Tech Stack:** Java, JavaFX, SQL/MySQL, Maven, FlutterFlow, Symfony, PHP, Stripe

PERSONAL PROJECTS

Homy | QR/NFC Based Pet Safety & Recovery Platform [🔗](#), (Web site) 2025

- Built a pet-recovery solution generating QR/NFC tags linked to pet profiles for fast reunification of lost pets.
 - Implemented profile management, QR generation, owner contact workflows, and location sharing for finders.
- Tech Stack:** Next.js, TypeScript, Tailwind CSS, shadcn/ui, PostgreSQL, Prisma, Nodemailer, QRCode.js

TECHNICAL SKILLS

- Programming Languages:** JavaScript, TypeScript, Java, Python, PHP, C/C++, C#
- Frameworks & Libraries:** React, Next.js, NestJS, Express, React Native, Angular, Symfony, Laravel, Spring Boot, .NET
- Databases:** PostgreSQL, MongoDB, MySQL, Redis
- Cloud & DevOps:** AWS (S3, SES), Docker, Git, Jenkins, PM2, Linux, Vercel, Netlify, OVH
- Tools & Platforms:** Jira, Trello, Slack, Linear, Figma, Sentry, Sanity, Arduino, Qt, N8N, GitHub, GitLab, Flutter Flow, Postman, Swagger

LANGUAGES

French: C1 (TCF) | **English:** C1